LUMAFLOW · RISK REGISTER

LumaFlow risk register and hazard analysis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	P8 mitigation as decided by Amish (O-4 (a), 2026-10-25)
0.1	2026-10-25	Amish Chadha	Kit 1.8 upgrade: project risks, preliminary hazard analysis and first-pass design FMEA (P3) for option A, updated for the independent review of 2026-10-25

LumaFlow risk register and hazard analysis

The top risk is a false sense of safety: a calculated dose is not a demonstrated one, and a household could trust LumaFlow's water when the liner reflects less, the water is cloudier or the firmware rule is wrong. The controls are the normally closed valve that stops the water on a low sensor reading, conservative dose mapping, and plain statements everywhere that LumaFlow is a research and educational prototype, not a certified device. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be found) to 5 (unlikely to be found).

> **Safety:** LumaFlow emits UV-C, holds mains water pressure next to electronics and is meant to make water safer to drink. It is a research and educational prototype, not a certified water treatment device. Nothing in this document is a test result; every rating is a paper estimate.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	Dose below target in the built reactor	Liner reflectance below 0.95, LED bin low, 275 nm response below 254 nm	PR-002 not met; more LEDs or less flow	3	4	Design lead	Margins stated (S3, S8, B9); measure reflectance and do biodosimetry at TRL 4
P2	No co-design partner	The extension programme is not approached or declines	Requirements stay unvalidated; no real UVT data	3	3	Project lead (Amish)	Draft first message exists (REVIEW 2026-10-02); second candidates in PLN
P3	Cost stays over the value-engineering target	Machined 316 cap, specialist liner, LED prices	USD 433 against USD 340	4	2	Design lead	Savings in DEC (plain PTFE, one machine-shop quote, LED quantity prices)
P4	Concept choice changes the project	Option B scores level with A	Redesign of every document if B is chosen	2	4	Project lead (Amish)	Decision O-1 put to Amish with both options argued (LMF-DDR-004)
P7	Dose margin thinner than the note's rule	275 nm response factor and LED derating (S3, S33)	PR-002 at risk; 37.5 mJ/cm² worst case	3	4	Design lead	Decision O-3 after the LED is chosen
P8	Reactor mixes more than assumed	Inlet jet and buoyancy (S34)	RED below the laminar value	3	4	Design lead	Tracer test at TRL 4 with the side inlet kept; perforated PTFE inlet ring ready (O-4 (a), decided by Amish 2026-10-25)
P5	Firmware dose rule is wrong or slow	Mapping and 1 s fail-safe exist only on paper	Untreated water passes during a fault	3	5	Electronics lead	PR-004 at risk; firmware sketch review and bench test at TRL 4
P6	Misuse of the open design as a certified product	A builder sells or relies on copies	Harm and reputational risk	2	5	Project lead	PR-019 labels and documents; CERN-OHL-S licence; no certification claims

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	UV-C exposure of eyes or skin	LED head removed or lid opened while powered; light leak at a seal or port	Photokeratitis, skin erythema	4	2	Metal and PTFE light path; lid reed switch; interlock loop in the LED head plug on a cable too short to clear the cap (PR-012); warning labels; UV-C eyewear for bench work	Low
H2	Under-treated water drunk	Cloudy water, fouled window, aged LEDs, sensor or firmware fault	Waterborne illness	5	2	Wall sensor with conservative proportional mapping; normally closed valve closes on low dose, fault or power loss (PR-004); pre-filter required; not a virus barrier (S5)	Medium (PR-004 at risk until tested)
H3	False sense of safety	User trusts a research prototype	Illness from chemical or viral contamination UV does not treat	5	3	PR-019 labelling; README and PRC safety notes; no certification claims	Medium
H4	Leak or burst under pressure	Window crack from a pressure spike; seal failure	Flooded cabinet; water on electronics	3	2	Window 6.18 MPa at 8 bar against 6.8 MPa; pressure limiter at 4 bar (PR-017); gasket seated at 2.4 MPa at 8 bar (S13); leak test before wiring	Low
H5	Electric shock	Mains in a wet cabinet	Injury	5	1	Only 24 V DC at the unit; certified adapter on a GFCI or RCD outlet above the water level (PR-013)	Low
H6	Hot surface or LED overheating	LEDs stuck on with no flow	Burn risk low; LED damage	2	2	Board sensor cuts current above 50 °C; stuck-on fault heads for 53 °C (CAL F6)	Low
H7	Material leaching into water	Wrong grade of plastic or rubber, UV-aged polymer	Contaminants in drinking water	3	2	Food-contact grades (PR-011); expanded PTFE in the light path (D-S1); acetal exposure to verify (S10)	Low to medium
H8	Bacterial regrowth after treatment	Water standing in the reactor and downstream tubing	Contaminated first draw	3	3	Idle LED pulse every 4 h; no residual disinfectant (stated); plate counts at TRL 4 (S7)	Medium

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	PTFE liner (line 2)	Reflect UV-C back into the water	Reflectance lower than 0.95 when wetted	Dose falls (45.2 mJ/ cm² at 0.80)	Grade, surface fouling	4	3	3	36	Supplier data and a measurement at TRL 4; the sensor sees a fouled liner as low signal
F2	UV-C LEDs (line 4)	Emit 60 mW each at 275 nm	Output low or aging	Dose falls	Bin, junction temperature, hours	4	3	2	24	Proportional mapping blames every loss on the LEDs (CAL C5); service-due segment on the bar
F3	Quartz window (line 3)	Pass UV-C, hold pressure	Crack	Leak into the LED head; loss of treatment	Pressure spike, uneven clamping	4	2	2	16	Pressure limiter (PR-017); ring screws in anti-seize at 0.21 N·m (D-S2); fuse and leak logic
F4	Window gasket (line 17)	Seal the window to the cap	Leak	Water in the LED head	Under-seated gasket, PTFE creep	3	2	3	18	Expanded PTFE seated at 1.8 MPa or more at 8 bar (S13); re-torque check 24 h after the first fill (build plan first checks)
F5	Wall photodiode (line 9)	Measure UV-C in the water	Reads high (stuck or saturated)	Low dose not caught	Electronics fault, stray light	5	2	4	40	Sensor-fail detection in firmware (PR-004); light-tight holder; reading checked on the idle pulse
F6	Flow sensor (line 8)	Confirm flow after start-up	No pulses at low flow	Valve shut as a fault, or water passes unmetered	Sensor below its range, fouled rotor	4	2	3	24	LEDs come on before the valve opens (D-S3); no pulse in 0.5 s shuts the valve; the dose estimate uses the measured flow
F11	Outlet pressure sensor (line 20)	Start the unit when the faucet opens	Stuck or drifting	Unit does not start (fails safe), or starts without demand	Sensor fault, air in the tee	2	2	2	8	Start-up check in the build plan; controller watches for a reading out of range
F7	Solenoid valve (line 11)	Stop water on alarm or	Sticks open	Untreated water on a fault	Debris, wear	5	2	4	40	Pre-filter; graft from option B (O-2): leak and stuck-valve check

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
		power loss								with the valve shut on each idle pulse
F8	LED head interlock (line 10)	Stop the LEDs with the head off	Loop shorted or bypassed	UV-C exposure during service	Wiring fault, tampering	4	1	3	12	Cable too short for the head to clear the cap while plugged in (model check); warning labels
F9	Acetal upper cap (line 7)	Hold the outlet and the tube end	Ages under UV-C passing through the PTFE	Cracking, leaching	PTFE transmittance at 275 nm	3	2	4	24	Measure at TRL 4 (S10); fallback 316 foil cup in the liner pocket
F10	Retaining ring screws (line 16)	Clamp the window stack	Over-torqued	Ring yields, washer creeps, window loads unevenly	0.5 N·m dry in the earlier plan; scatter of the dry nut factor	3	2	3	18	Anti-seize and 0.21 N·m with a torque screwdriver, ring in 316 (D-S2)